

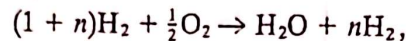
tion temperature would be approximately 3800 K and the product composition quite different from the product resulting from complete combustion.

To demonstrate the striking effect of dissociation on rocket performance, Fig. 12.1 shows the results of calculations of specific impulse as a function of fuel-oxidant ratio for the two extremes:

1. Complete combustion; that is, either all the fuel or all the oxidant consumed.
2. Equilibrium; that is, combustion product composition determined by thermodynamic equilibrium in the combustion chamber.

For both these cases we have assumed that there is no chemical reaction in the nozzle.

For complete combustion we can easily estimate the variation of specific impulse with fuel-oxidant ratio. With excess hydrogen the reaction is



where n can have any positive value. From this and Eq. (11.4) we can see that

$$I_{sp} = \frac{u_e}{g_c} = \frac{1}{g_c} \sqrt{2 \left(c_p T_{01} + \frac{\bar{Q}_R}{18 + 2n} \right) \left[1 - \left(\frac{p_e}{p_0} \right)^{(\gamma-1)/\gamma} \right]}, \quad (12.1a)$$

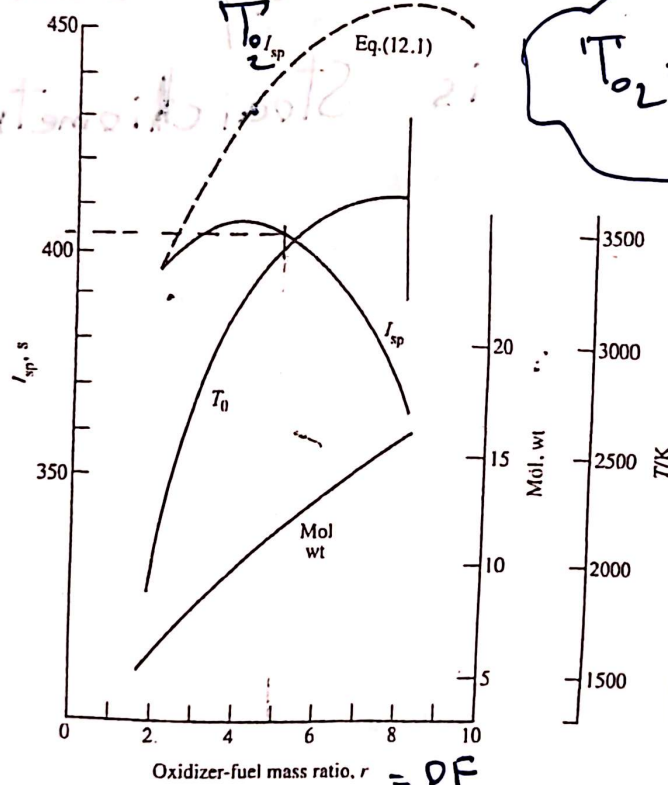


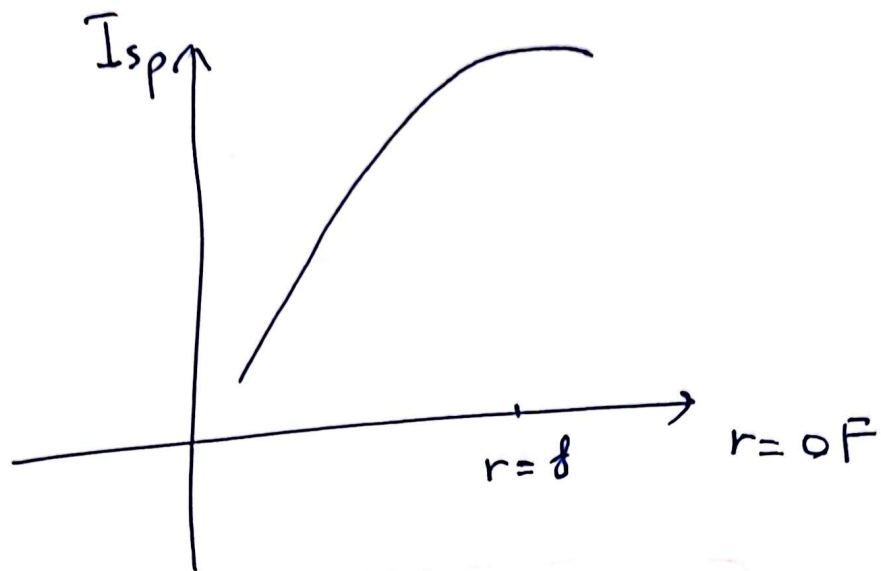
FIGURE 12.1 $\text{H}_2\text{-O}_2$; shifting equilibrium expansion from 6.89 MPa (1000 psia) to 0.1 MPa (14.7 psia); reactants at 298 K.

$$T_{02} = T_{01} + \frac{\bar{Q}_R}{c_p \bar{M}_m}$$

$r > r_{st}$
 $M_m \uparrow$ on the
 product part
 unburnt fuel

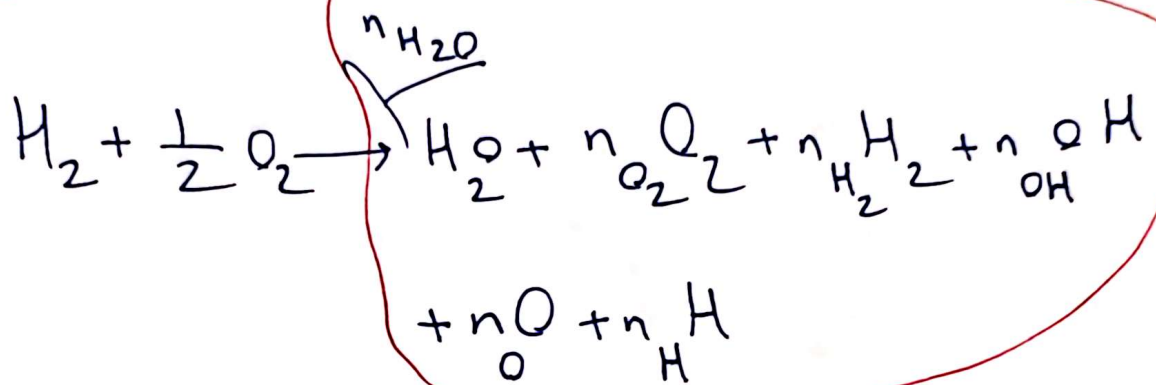
$r < r_{st}$
 $Q_R \downarrow$ as the

as not all the
 energy of fuel was
 released.

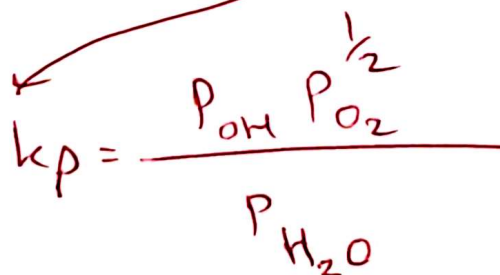
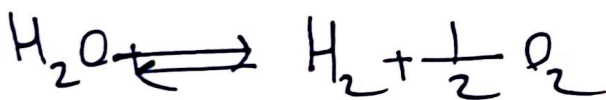
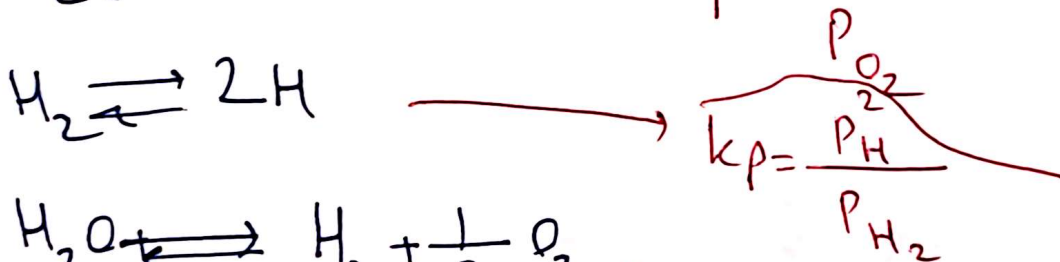


Max I_{sp} occurs when combustion is Stoichiometric

Mixture
after
combustion



on the right, we have these elementary reaction



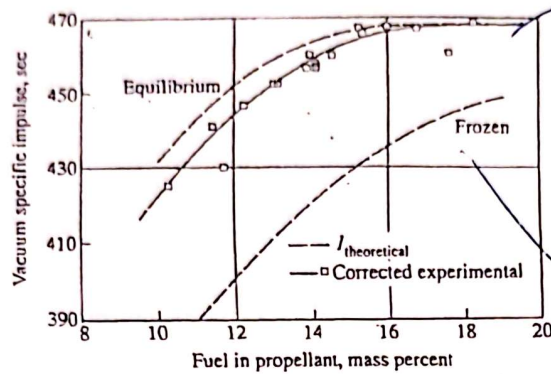


FIGURE 12.5 "Corrected" specific impulse of hydrogen-oxygen rocket. Combustion pressures 0.27, 0.14, 0.5 MPa. (Courtesy Olson [4].)

Combustion & reaction occurs completely more energy is realised

Not complete reaction lower energy realises

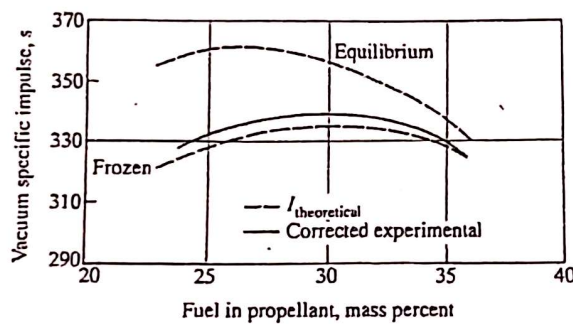


FIGURE 12.6 "Corrected" specific impulse of JP4-oxygen rocket. Combustion pressure 4 MPa. (Courtesy Olson [4].)

Reactions continue in the nozzle

Shifting equilibrium expansions and frozen expansion:

In the former, the reactions take place so fast that equilibrium prevails at every nozzle station. "shifting" because the equilibrium (composition) is changing due to changing pressure and temperature. In the latter, the reactions are so slow no appreciable composition change occurs; that is, the composition is chemically frozen.

The nature of the nozzle expansions is determined by the relative magnitudes of the flow time and the reaction time.

$\tau_{\text{flow}} \sim \frac{L}{u}$, τ_{reac} depends on reaction kinetics, higher

The temperature and pressure, smaller the τ_{reac}

Near the nozzle entrance, T and p are high and u is small.

Therefore, τ_{reac} is small and τ_{flow} is large.

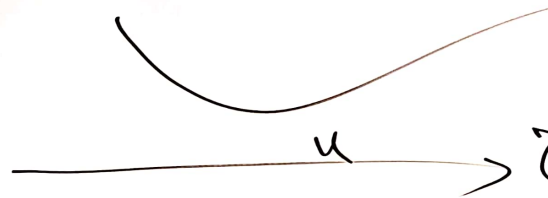
$$\tau_{\text{reac}} \ll \tau_{\text{flow}} \Rightarrow \text{equilibrium}$$

Near the nozzle exit, T and p are low, and u is large

Therefore, τ_{reac} is high and τ_{flow} is small

$$\tau_{\text{reac}} \gg \tau_{\text{flow}} \Rightarrow \text{chemically frozen}$$

A rough approximation to get satisfactory results is to assume shifting equilibrium from the nozzle entrance to the nozzle throat, and frozen flow thereafter.



$$\tau_{flow} \sim \frac{L}{u}$$

$$\tau_{reaction} \begin{cases} P \uparrow \\ P \uparrow \end{cases} \rightarrow \tau_{rea} \downarrow$$



$$\tau_{flow} \gg \tau_{reaction}$$

